

PRACTICE MIDTERM II (with solutions, along with 2 solved book problems)

Please show your work and write only in **pen**. Calculators and notes are forbidden. Only 1 brain (your own) allowed per exam. Please also note that there are **13** problems, some of which have multiple parts.

1: Please find the characteristic polynomial of $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 2 & -4 \\ 2 & 3 & 5 \end{bmatrix}$.

Done in class on Thursday, Oct. 26, 2017.

2: Please write the solution to the linear system $\begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ as a vector of linear

functions in 2 parameters. Then describe how you can find 2 vectors in $\mathbb{R}^{5 \times 1}$ whose linear combinations describe all solutions of the preceding linear system.

The first thing to observe is that the system is already in reduced row echelon form. In particular, the leading 1s are boxed below:

$$\begin{bmatrix} \boxed{1} & 0 & 0 & 0 & 2 \\ 0 & 0 & \boxed{1} & 0 & 3 \\ 0 & 0 & 0 & \boxed{1} & 5 \end{bmatrix}$$

(Note that leading 1s are always the first nonzero entries in their respective rows (going left to right), and there are no nonzero entries above or below a leading 1.) Since the leading 1s are in columns 1, 3, and 4, this means that x_1 , x_3 , and x_4 are the leading variables here, and can be expressed as a function of the remaining (free) variables x_2 and x_5 . In particular, it's easy to see that

$$\begin{aligned} x_1 &= -2x_5 \\ x_3 &= -3x_5 \\ x_4 &= -5x_5, \end{aligned}$$

where x_2 and x_5 are free (i.e., allowed to take on any value). In other words, the set of solutions of our linear system is described completely by:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2\beta \\ \alpha \\ -3\beta \\ -5\beta \\ \beta \end{bmatrix},$$

where $\alpha, \beta \in \mathbb{R}$ are arbitrary (if one wants to work over the real numbers).

Alternatively, we can simply write all solutions in the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \alpha \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -2 \\ 0 \\ -3 \\ -5 \\ 1 \end{bmatrix}.$$

Note in particular that each such solution is *uniquely* determined by (α, β) , if one looks at the second and fifth coordinates. So we can in fact say that the two preceding vectors

define a *basis* for our solution space. We'll say more about the definition of basis in lecture after Midterm II...

3: Please write the solution to the linear system $\begin{bmatrix} 1 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 1 & -13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ as a vector of linear functions in 3 parameters. Then describe how you can find 3 vectors in $\mathbb{R}^{5 \times 1}$ whose linear combinations describe all solutions of the preceding linear system.

Proceeding as in our solution to Problem #2, observe that our system is already in reduced row echelon form, with leading 1s in the first and fourth columns. So then x_1 and x_4 are the leading variables here, and can be expressed as a function of the remaining (free) variables x_2 , x_3 , and x_5 . In particular, it's easy to see that

$$\begin{aligned} x_1 &= -7x_5 \\ x_4 &= 13x_5, \end{aligned}$$

where x_2 , x_3 , and x_5 are free (i.e., allowed to take on any value). In other words, the set of solutions of our linear system is described completely by:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -7\gamma \\ \alpha \\ \beta \\ 13\gamma \\ \gamma \end{bmatrix},$$

where $\alpha, \beta, \gamma \in \mathbb{R}$ are arbitrary (if one wants to work over the real numbers).

Alternatively, we can simply write all solutions in the form $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \alpha \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} -7 \\ 0 \\ 0 \\ 13 \\ 1 \end{bmatrix}.$

Note in particular that each such solution is *uniquely* determined by (α, β, γ) , if one looks at the second, third, and fifth coordinates. So we can in fact say that the two preceding vectors define a *basis* for our solution space.

4: Please write the solution to the linear system $\begin{bmatrix} 1 & -12 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ as a vector of linear functions in 3 parameters. Then describe how you can find 3 vectors in $\mathbb{R}^{5 \times 1}$ whose linear combinations describe all solutions of the preceding linear system.

Proceeding as in our solution to Problem #3, observe that our system is already in reduced row echelon form, with leading 1s in the first and fourth columns. So then x_1 and x_4 are the leading variables here, and can be expressed as a function of the remaining (free) variables x_2 , x_3 , and x_5 . In particular, it's easy to see that

$$\begin{aligned} x_1 &= 12x_2 - x_5 \\ x_4 &= 2x_5, \end{aligned}$$

where x_2 , x_3 , and x_5 are free (i.e., allowed to take on any value). In other words, the set of solutions of our linear system is described completely by:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 12\alpha - \gamma \\ \alpha \\ \beta \\ 2\gamma \\ \gamma \end{bmatrix},$$

where $\alpha, \beta, \gamma \in \mathbb{R}$ are arbitrary (if one wants to work over the real numbers).

Alternatively, we can simply write all solutions in the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} -1 \\ 0 \\ 0 \\ 2 \\ 1 \end{bmatrix}.$$

Note in particular that each such solution is *uniquely* determined by (α, β, γ) , if one looks at the second, third, and fifth coordinates. So we can in fact say that the two preceding vectors define a *basis* for our solution space.

5: Assume $\theta \in \mathbb{R}$. Please find the eigenvalues (and corresponding eigenvectors) for $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, as a function of θ .

Using the trace formula, we easily see that the characteristic polynomial here is (up to sign) $\lambda^2 - 2\cos(\theta)\lambda + (\cos^2(\theta) + \sin^2(\theta)) = \lambda^2 - 2\cos(\theta)\lambda + 1$. So by the quadratic formula, the roots are exactly $\cos(\theta) \pm \frac{\sqrt{4\cos^2(\theta)-4}}{2} = \cos(\theta) \pm \frac{2\sqrt{-(1-\cos^2(\theta))}}{2} = \cos(\theta) \pm \sqrt{-\sin^2(\theta)} = \cos(\theta) \pm \sin(\theta)\sqrt{-1} = e^{\pm\theta\sqrt{-1}}$. (The last equality follows from the Euler's famous formula from 1748: $e^{\theta\sqrt{-1}} = \cos(\theta) + \sin(\theta)\sqrt{-1}$.) So the eigenvalues are exactly $\boxed{\{e^{-\theta\sqrt{-1}}, e^{\theta\sqrt{-1}}\}}$.

Let us now compute the corresponding eigenvalues...

$(\lambda = e^{-\theta\sqrt{-1}})$: We need to see what vectors v satisfy $A - \lambda I$ in our setting. So this boils down to finding a vector $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ such that $\begin{bmatrix} \cos \theta - e^{-\theta\sqrt{-1}} & -\sin \theta \\ \sin \theta & \cos \theta - e^{-\theta\sqrt{-1}} \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. So this means we solve

$$\begin{bmatrix} \cos(\theta) - (\cos(\theta) - \sin(\theta)\sqrt{-1}) & -\sin(\theta) \\ \sin(\theta) & \cos \theta - (\cos(\theta) - \sin(\theta)\sqrt{-1}) \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

or

$$\begin{bmatrix} \sin(\theta)\sqrt{-1} & -\sin(\theta) \\ \sin(\theta) & \sin(\theta)\sqrt{-1} \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Clearly then, $v = \begin{bmatrix} 1 \\ \sqrt{-1} \end{bmatrix}$ works, and is thus the eigenvector corresponding to the eigenvalue $\lambda = e^{-\theta\sqrt{-1}}$.

$(\lambda = e^{\theta\sqrt{-1}})$: Similar to our last calculation, we need to find a vector $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ such that

$$\begin{bmatrix} \cos \theta - e^{\theta\sqrt{-1}} & -\sin \theta \\ \sin \theta & \cos \theta - e^{\theta\sqrt{-1}} \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \text{ So this means we solve}$$

$$\begin{bmatrix} \cos(\theta) - (\cos(\theta) + \sin(\theta)\sqrt{-1}) & -\sin(\theta) \\ \sin(\theta) & \cos \theta - (\cos(\theta) + \sin(\theta)\sqrt{-1}) \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

or

$$\begin{bmatrix} -\sin(\theta)\sqrt{-1} & -\sin(\theta) \\ \sin(\theta) & -\sin(\theta)\sqrt{-1} \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Clearly then, $v = \begin{bmatrix} 1 \\ -\sqrt{-1} \end{bmatrix}$ works, and is thus the eigenvector corresponding to the eigenvalue $\lambda = e^{\theta\sqrt{-1}}$.

Note that here, we have worked with complex matrices, complex eigenvalues, and complex eigenvectors.

6: (a) Please find a 2×2 matrix, that is *not* the zero matrix, whose only eigenvalue is 0.
 (b) Please prove that $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ is *not* diagonalizable, i.e., there is *no* invertible 2×2 matrix S with $S \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} S^{-1}$ diagonal. **Hint:** First find the eigenvalues $\{\lambda_1, \lambda_2\}$ of $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$. Then prove that the identity $S \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} S$ is impossible to attain.

(a) There are infinitely many examples. One such is $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$: It is easy to see that the characteristic polynomial here is λ^2 , and thus 0 is the only eigenvalue. However, our matrix is clearly not the zero matrix.

(b) We omit the easy calculation that 3 is the only eigenvalue, and go directly to the proof of non-diagonalizability: We assume we *can* diagonalize, derive a contradiction, and thus conclude that diagonalization is *not* possible here.

More precisely, if there were an $S = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $S \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} S^{-1}$ diagonal, then the (1,2) and

(2,1) entries of $S \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} S^{-1}$ would both have to be 0. So let's find these entries:

$$S \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} S^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \cdot \frac{1}{ad-bc} = \begin{bmatrix} 3a & a+3b \\ 3c & c+3d \end{bmatrix} \begin{bmatrix} * & -3ab+a^2+3ab \\ 3cd-c^2-3cd & * \end{bmatrix} \cdot \frac{1}{ad-bc}.$$

(The “*” indicates entries we don't care about for now.) So the entries that must vanish are $\frac{a^2}{ad-bc}$ and $\frac{-c^2}{ad-bc}$, and we therefore must have $a=c=0$. So then S must have a 0 column. But then, this means that S is non-invertible, and thus we derive a contradiction (since we need an invertible S in order to have diagonalizability!).

7: Please express $\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}^{128}$ in the form $\begin{bmatrix} a\alpha^{128} + b\beta^{128} & c\alpha^{128} + d\beta^{128} \\ r\alpha^{128} + s\beta^{128} & u\alpha^{128} + v\beta^{128} \end{bmatrix}$ (for some explicit constants $a, b, c, d, r, s, u, v, \alpha, \beta$).

Done in class on Thursday, Oct. 26, 2017.

8: Please express $e^{\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}}$ in the form $\begin{bmatrix} ae^\alpha + be^\beta & ce^\alpha + de^\beta \\ re^\alpha + se^\beta & ue^\alpha + ve^\beta \end{bmatrix}$ (for some explicit constants $a, b, c, d, r, s, u, v, \alpha, \beta$).

Done in class on Thursday, Oct. 26, 2017.

9: Suppose $A \in \mathbb{R}^{2 \times 2}$ has trace 11 and determinant 18. What are the eigenvalues of A ?

From our trace formula, this means that the characteristic polynomial is $\lambda^2 - 11\lambda + 18$. This obviously factors as $(\lambda - 9)(\lambda - 2)$, so the eigenvalues are $\boxed{\{2, 9\}}$.

10: Suppose $A \in \mathbb{R}^{n \times n}$ is invertible and has a nonzero eigenvalue λ . Please prove that A^{-1} has $\frac{1}{\lambda}$ as an eigenvalue.

Directly from the definition, we have that there must be a nonzero vector $v \in \mathbb{R}^n$ with $Av = \lambda v$.

Since λ is nonzero, we can divide both sides by λ to get

$$\frac{1}{\lambda} Av = v.$$

Since A is invertible, we multiply both sides on the left by A^{-1} to get

$$\frac{1}{\lambda} A^{-1} Av = A^{-1} v.$$

So we get

$$A^{-1} v = \frac{1}{\lambda} v,$$

which is the definition of A^{-1} having eigenvalue $\frac{1}{\lambda}$.

11:

(a) Please prove that if $A \in \mathbb{C}^{n \times n}$ is a matrix satisfying $A^2 = A$ then any eigenvalue of A must be 0 or 1.

(b) Please prove that if $A \in \mathbb{C}^{n \times n}$ satisfies $A^k = \mathbf{O}$ (the zero matrix) for some integer k then all the eigenvalues of A are 0.

(c) Please find the first 4 powers of $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) For basic eigenvalue proofs, it is usually best to be as direct as possible: Assuming $Av = \lambda v$ for some eigenpair (v, λ) , observe then that $AAv = \lambda Av$. Now, we evaluate AAv two different ways: On the one hand, $AAv = A(\lambda v) = \lambda Av = \lambda \lambda v = \lambda^2 v$. On the other hand, since $A^2 = A$, we must also have $AAv = Av = \lambda v$. So then, $\lambda^2 v = \lambda v$.

Now, if $\lambda = 0$, then we arrive at what we were trying to prove. So assume $\lambda \neq 0$. Then, since v is a nonzero vector, we must have $\lambda^2 = \lambda$, i.e., $\lambda = 1$ (since we've just assumed $\lambda \neq 0$). So $\lambda \in \{0, 1\}$ and we are done.

(b) Here is a fast proof by contradiction: Suppose A has a *nonzero* eigenvalue λ . Then there must be a nonzero vector $v \in \mathbb{C}^n$ with $Av = \lambda v$. Multiplying both sides on the left by A^{k-1} we then get $A^k v = A^{k-1} Av = A^{k-1} \lambda v = \lambda A^{k-1} v = \lambda^2 A^{k-2} v = \dots$ etc. So we get $A^k v = \lambda^k v$ by induction. But A^k is the zero matrix, so we get $A^k v$ equal to the zero vector, and thus $\lambda^k v$ is the zero vector --- a contradiction, since λ was assumed to be nonzero and v was assumed to be a nonzero vector! So we must have that all eigenvalues of A are 0.

(c) Going through the usual motions, one easily gets $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}^1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}^3 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}^4 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

12:

Suppose $A \in \mathbb{C}^{5 \times 5}$, I is the 5×5 identity matrix, and A has eigenvalues 4, 2, 7, 9, and -1 . Please find the eigenvalues of $A - \sqrt{2}I$ and prove why your answer is correct.

Done in class on Thursday, Oct. 26, 2017.

13:

Suppose $M \in \mathbb{R}^{3 \times 3}$ is a matrix satisfying $M \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$ and $M \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

(a) Please find two distinct eigenvalues of M . (b) Please find two distinct eigenvectors of M .

Done in class on Thursday, Oct. 26, 2017.

Problem 2 from Sec. 6.1 of Leon 8th ed.:

Show that the eigenvalues of a triangular matrix are the diagonal elements of the matrix.

First note that we need to consider two cases: upper-triangular matrices and lower-triangular matrices. However, it's enough to consider just one case, say the upper-triangular case. This is because we can simply apply matrix transpose to go from one proven case to the other, and determinants are invariant with respect to transpose, i.e., $\det(A) = \det(A^T)$.

Now, to prove the upper-triangular case, consider any $A = [a_{i,j}] \in \mathbb{C}^{n \times n}$ and its characteristic polynomial $p(\lambda) := \det(A - \lambda I)$: The roots of p are exactly the eigenvalues of A . Now, recall that the determinant of any upper-triangular matrix is the product of its diagonal entries, and observe that $A - \lambda I$ is upper-triangular. So $p(\lambda) = (\lambda - a_{1,1})(\lambda - a_{2,2}) \cdots (\lambda - a_{n,n})$. So the eigenvalues are exactly $a_{1,1}, a_{2,2}, \dots, a_{n,n}$ and we are done.

Problem 33 from Sec. 6.1 of Leon 8th ed.: Let A and B be $n \times n$ matrices (over some field). Show that

(a) If λ is a nonzero eigenvalue of AB then it is also an eigenvalue of BA .

(b) If $\lambda=0$ is an eigenvalue of AB then $\lambda=0$ is also an eigenvalue of BA .

(a) By definition, there must be a nonzero vector v satisfying $ABv=\lambda v$. Multiplying on the left by B , we then get $BABv=B\lambda v$. But this is the same as

$$(\star) \quad (BA)(Bv)=\lambda(Bv).$$

Now since, λv is not the zero vector, ABv is also a nonzero vector. In particular, Bv can't be the zero vector (for otherwise we would have $ABv=A(Bv)$ being the zero vector). So $(\star)$ tells us that Bv is an eigenvector of BA corresponding to the eigenvalue λ , and we are done.

(b) The trick here is to use prove by contradiction, because the argument for (a) can't be adapted for $\lambda=0$: First recall that for a product of two matrices ST to be invertible, we must have S and T invertible as well. So, to derive a contradiction, assume BA is invertible. Clearly then, B must be invertible as well. Now observe that, since $ABv=\mathbf{0}$, we can multiply both sides on the left by A^{-1} to obtain $Bv=\mathbf{0}$. But this is impossible!: $Bv=\mathbf{0}$ implies that v is a nonzero right null vector of B . Invertible matrices can't have such vectors, so we are done. We must have BA *non*-invertible, and thus 0 is an eigenvalue of BA .